

GRAYSON W. WHITE

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EDUCATION

Ph.D. Forestry, *Michigan State University*, East Lansing, MI. 2025

- Dissertation: *Methods for improved small area estimation in forest inventory*
- Advisor: Dr. Andrew Finley

M.S. Statistics, *Michigan State University*, East Lansing, MI. 2025

B.A. Mathematics (concentration Statistics), *Reed College*, Portland, OR. 2021

- Thesis: *A hierarchical Bayesian approach to small area estimation of forest attributes*
- Thesis advisor: Dr. Kelly McConville

ACADEMIC POSITIONS

Visiting Assistant Professor of Statistics, *Reed College*, Portland, OR. Aug 2025 – Present

- Courses taught: Math/Stat 141: Introduction to Probability and Statistics ([Fall 2025](#)), and Math/Stat 241: Data Science ([Spring 2026](#), [Fall 2026](#)), Stat 243: Statistical Learning ([Fall 2026](#)).

Undergraduate Forestry Data Science Lab Co-Director Summer 2022 – Present

- Co-director of the [Undergraduate Forestry Data Science Lab](#), with Dr. Kelly McConville and Dr. George Gaines. Focuses include developing projects, mentoring students, and improving understanding of our forests through data science.
- Past research has occurred at Harvard University and Bucknell University. The summer 2026 research program occurred on Reed College campus.

Instructor, *Michigan State University*, East Lansing, MI. Fall 2024

- Course taught: [Forestry 128: Practical Computing & Data Science Tools](#).

Harvard Fellow, *Harvard University*, Cambridge, MA. Summer 2022 and 2023

- Research fellow in the Department of Statistics, Harvard University, conducting research focused on zero-inflated modeling of forest attributes and accurately simulating forest structure in artificial populations with Dr. Kelly McConville.

Graduate Research Assistant, *Michigan State University*, East Lansing, MI. Aug 2022 – Dec 2025

- Graduate research assistant position in Dr. Andrew Finley's Geospatial Lab, focusing on small area estimation of forest attributes, hierarchical Bayesian geospatial statistical modeling, and forest inventory estimation.

Graduate Teaching Assistant, *Michigan State University*, East Lansing, MI. Spring 2023

- Courses: *Forestry 128: Practical Computing & Data Science Tools* and *Forestry 372: Ecological Monitoring and Data Analysis*

Research Fellow, *Reed College Forestry Data Science Lab*, Portland, OR.

Summer 2021

- Prepared and submitted manuscript titled *Hierarchical Bayesian Small Area Estimation Using Weakly Informative Priors in Ecologically Homogeneous Areas of the Interior Western Forests to Frontiers in Forests and Global Change*.

Research Fellow, *Data Science for the Public Good*, Oregon State University.

Summer 2020

- Developed forecasting tools for cost analysis of water and wastewater facilities in Oregon small towns and cities using Bayesian methods.

Undergraduate Teaching Assistant, *Reed College*, Portland, OR.

Jan 2020 – May 2021

- Courses: *Math 392: Mathematical Statistics*, *Math 241: Data Science*, *Math 141: Intro to Probability and Statistics*, and *Econ 311: Survey of Econometric Methods*.

PUBLICATIONS

* Indicates an undergraduate student.

Manuscripts under review

White G., McConville K., and Schumacher C. (2026+). *A penalized logistic generalized regression estimator*. [arXiv preprint](#). Submitted to the Journal of Survey Statistics and Methodology.

Stewardson D.*, **White G.**, and Groce A. (2026+). *Differentially Private Average Treatment Effect Estimation by Propensity Score Blocking*. [arXiv preprint](#). Submitted to the 2027 AAAI Conference on Artificial Intelligence.

Peer-reviewed publications

White G., Finley A., Yamamoto J., Green J., Frescino T., Andersen H., and Domke G. (2026). *Hierarchical models for small area estimation using zero-inflated forest inventory variables: comparison and implementation*. Canadian Journal of Forest Research.

White G., Wieczorek J., Cody Z.*, Tan E.*, Chistolini J.*, McConville K., Frescino T., and Moisen G. (2025). *A method for empirically assessing small area estimators via bootstrap-weighted k-Nearest-Neighbor artificial populations, with applications to forest inventory*. *Forestry: An International Journal of Forest Research*.

White G., Yamamoto J.*, Elsyad D.*, Schmitt J.*, Korsgaard N.*, Hu K., Frescino T., Gaines G., and McConville K. (2025). *Small area estimation of forest biomass via a two-stage model for continuous zero-inflated data*. Canadian Journal of Forest Research.

Emick E., Babcock C., **White G.**, Hudak A., Domke G., and Finley A. (2023). *An approach to estimating forest biomass while quantifying estimate uncertainty and correcting bias in machine learning maps*. Remote Sensing of Environment.

Frescino T., Moisen G., Patterson P., Toney C., and **White G.** (2023). *FIESTA: A Forest Inventory Estimation and Analysis R Package*. Ecography.

Frescino T., McConville K., **White G.**, Toney C., and Moisen G. (2022). *Small Area Estimates for National Applications: A Database to Dashboard Strategy Using FIESTA*. Frontiers in Forests and Global Change.

White G., McConville K., Moisen G., and Frescino T. (2021). *Hierarchical Bayesian Small Area Estimation Using Weakly Informative Priors in Ecologically Homogeneous Areas of Interior Western Forests*. Frontiers in Forests and Global Change.

Statistical Software: R Packages

Russell L.*, Wu W.*, **White G.**, McConville K., and Gaines G. (2026). *basal*: Bayesian Small Area Estimation Library. Author.

White G., Frick H., and Couch S. (2024). *forested*: Forest Attributes in Washington State. Author.

White G., Wieczorek J., Frescino T., and McConville K. (2024). *kbaabb*: A kNN-based approximation to the approximate Bayesian bootstrap. Author, maintainer.

Yamamoto J.*, Elsyad D.*, **White G.**, Schmitt J.*, Korsgaard N.*, McConville K., and Hu K. (2024). *saeczi*: Small Area Estimation for Continuous Zero Inflated Data. Author.

McConville K., Yamamoto J., Tang B., Zhu G., and **White G.** (2024). *mase*: Model-Assisted Survey Estimators. Author.

Frescino T., Moisen G., Patterson P., Toney C., **White G.**, and Yamamoto J. (2022). *FIESTA*: Forest Inventory Estimation and Analysis. Author, maintainer.

Frescino T., Toney C., **White G.**, and Yamamoto J. (2022). *FIESTAutils*: Utility Functions for Forest Inventory Estimation and Analysis. Author, maintainer.

White G. (2020). *ggglm*: Grammar of Graphics for Linear Model Diagnostic Plots. Official *ggplot2* extension. Author, maintainer.

INDUSTRY POSITIONS

Data Science Educator, *RStudio PBC (now Posit PBC)*, Remote.

Jan 2022 – June 2022

- Write instructional materials and develop content for R users including tutorials, code recipes, practice exercises, datasets, and large-scale data science projects.

- Lead and mentor groups of adult learners through data science projects as they complete apprenticeships with R.

Data Scientist, Redcastle Resources Inc., Ogden, UT.

July 2021 – Dec 2025

- Full-time position (July 2021 - May 2022, part-time May 2022 - December 2025) contracting with the USDA Forest Service Forest Inventory and Analysis (FIA) program. Primary projects include development of the *FIESTA* R package, small area estimation research, model-assisted estimation research, and R *shiny* dashboard development.

FUNDING ACQUISITION: GRANTS

- USDA Forest Service, Rocky Mountain Research Station, FIA Program. *Exploring applications of differential privacy methods in the Forest Inventory and Analysis Program*. (2026-2027). \$70,000. PI. As faculty.
- USDA Forest Service, Rocky Mountain Research Station, FIA Program. *Undergraduate Forestry Data Science: Summer 2026*. (2026). \$19,343. PI. As faculty.
- National Council for Air and Stream Improvement, Inc. *A national application of k nearest neighbor bootstrap weighted artificial populations for formal assessment of small area estimators*. (2024-2025). \$50,000. Co-PI. As graduate student.
- National Council for Air and Stream Improvement, Inc. *Creating testbed populations for PSAE simulation studies*. (2022-2023). \$20,000. Co-PI. As graduate student.
- USDA Forest Service, Rocky Mountain Research Station, FIA Program. *Improving small area estimation methodologies for forest inventories*. (2022-2025). \$75,000. Co-PI. As graduate student.

PRESENTATIONS

Conferences, invited

(Upcoming presentation). *A penalized logistic generalized regression estimator*, Forest Inventory and Analysis Science Symposium 2026, Remote, October 2026.

Zero-inflated models for small area estimation in forest inventory: a comparison of frequentist and Bayesian methods, American Statistical Association Oregon Chapter Fall 2025 Meeting, Portland, Oregon, United States, November 2025.

Small area estimation in practice: why you should be a Bayesian about it, Forest Inventory and Analysis Science Symposium 2024, Remote, November 2024.

Improving Forest Inventory Small Area Estimates Through Ecological Borrowing and Hierarchical Bayesian Methods, Forest Inventory and Analysis Science Stakeholder Meeting 2022, Remote, November 2022.

Conferences, contributed

gglm: A tool for teaching linear model diagnostics while adhering to the grammar of graphics, Joint Statistical Meetings 2026, Boston, Massachusetts, United States, August 2026. Topic-contributed session.

Hierarchical models for small area estimation using zero-inflated forest inventory variables: comparison and implementation, International Society for Bayesian Analysis World Meeting, Nagoya, Japan, July 2026. Poster presentation.

Teaching and Learning via Undergraduate Research: Notes from the Harvard Forestry Data Science Lab, Joint Statistical Meetings 2024, Portland, Oregon, United States, August 2024. Poster presentation.

Assessing small area estimates via bootstrap-weighted k-Nearest-Neighbor artificial populations, Joint Statistical Meetings 2024, Portland, Oregon, United States, August 2024. Contributed presentation.

A spatial small area model for predicting average forest biomass using fine resolution geospatial data, Joint Statistical Meetings 2023, Toronto, Ontario, Canada, August 2023. Contributed presentation.

How Do We Assess the Performance of Our Small Area Estimators?, FIA Science Stakeholder Meeting 2022, Remote, November 2022. Contributed presentation.

Hierarchical Bayesian Small Area Estimation Using Weakly Informative Priors in the Interior Western United States, SAE 2022: Small Area Estimation, Surveys and Data Science, University of Maryland, College Park, Maryland, United States, May 2022. Contributed presentation.

Guest lectures

Improving forest inventory estimates in small regions with modern methodologies and novel simulation studies, EFB 439: Forest Health Monitoring, SUNY ESF, Remote, May 2026.

The FIA Program: A Statistician's Perspective, EFB 439: Forest Health Monitoring, SUNY ESF, Remote, May 2026.

MENTORING

Senior Theses

Schwa Yeleti (Mathematics '27, Reed College)

- Thesis (in progress): *Model-assisted change estimation for forest inventory attributes*.

Jacob Bainton (Mathematics—Computer Science '27, Reed College)

- Thesis (in progress): *Implementation of a multi-label tree edit distance algorithm*.
- Co-advised with Dr. Harper Knittel

Ida Landecker (Biology—Environmental Studies '27, Reed College)

- Thesis (in progress): *A spatio-temporal analysis of thermal refuges.*
- Co-advised with Dr. Sam Fey

Nat Lincoln (Mathematics with a Concentration in Statistics '26, Reed College)

- Thesis: *Measuring Greenspace Accessibility in Portland, Oregon*

Duncan Stewardson (Mathematics—Computer Science '26, Reed College)

- Thesis: *An Exploration of Differentially Private Average Treatment Effect Estimation*
- Co-advised with Dr. Adam Groce

Forestry Data Science Research Fellows

Summer 2026:

- Brendan Fingal (Mathematics with a Concentration in Statistics '28, Reed College)
- Leland Russell (Mathematics with a Concentration in Statistics '27, Reed College)
- Sam Schroeder (Electrical and Electronics Engineering '29, Bucknell University)
- Christian West-McCauley (Geography '27, Bucknell University)
- Wystan Wu (Mathematics with a Concentration in Statistics '27, Reed College)

Summer 2025:

- Calvin Bailey (Data Analytics '25, Denison University)
- Rebecca Beneroff (Biology BA '25, Biology MA '27, Bucknell University)
- Austin Carroll (Statistics '26, Bucknell University)
- Celine Chang (Applied Mathematics '27, Bucknell University)
- Odilon Ligan (Computer Science & Engineering '27, Bucknell University)
- Will Lindquist (Statistics '27, Bucknell University)
- Jean Marie Ngabonziza (Computer Science & Engineering '26, Bucknell University)
- Fayrene Nguyen (Business Analytics '27, Bucknell University)

Summer 2023:

- Dinan Elsyad (Statistics '25, Harvard University)
- Lal Kablan (Psychology and Economics '25, Harvard University)
- Niels Korsgaard (Statistics '25, Harvard University)
- Thor Reimann (Religion, Environmental Science, and Public Policy '25, Harvard University)

Summer 2022:

- Asteria Chilambo (Mathematics '23, Harvard University)
- Julian Schmitt (Applied Mathematics '23, Harvard University)
- Jing Shang (Economics '23, Fudan University)
- Maxwell VanLandschoot (Economics '22, Reed College)
- Josh Yamamoto (Mathematics with a Concentration in Statistics '23, Reed College)

Post-Baccalaureate Students

Duncan Stewardson (Summer 2026, Fall 2026 - Spring 2027)

- Project (Summer 2026): Differentially private average treatment effect estimation by propensity score blocking
- Project (Fall 2026 - Spring 2027): Exploring the applications of differential privacy for the FIA Program

PROFESSIONAL SERVICE

Session chair, session titled “Modern Record Linkage Techniques for a Complex and Ever-changing World.” International Society for Bayesian Analysis World Meeting. Nagoya, Japan, July 2026.

Session organizer, session titled “Data Science Education: Innovations and Techniques.” American Mathematical Society Spring Western Sectional Meeting. Boise, Idaho, United States, March 2026. Co-organized with Dr. Eric Friedlander.

Peer Reviewer, Journal of the Royal Statistical Society: Series A

Panelist, Research Panel for the National Partnership for Small Area Estimation

Panelist, Development Panel for the National Partnership for Small Area Estimation

Discussant, session titled “Small Area Estimation in Forestry Inventory”. SAE 2022: Small Area Estimation, Surveys and Data Science, University of Maryland, College Park, MD, May 2022.

PROFESSIONAL AFFILIATIONS

Member (since 2022), American Statistical Association

Member (since 2022), International Society for Bayesian Analysis

Member (since 2025), Isolated Statisticians

Member (since 2026), American Mathematical Society

Invitational Member (since 2022), National Partnership for Small Area Estimation